

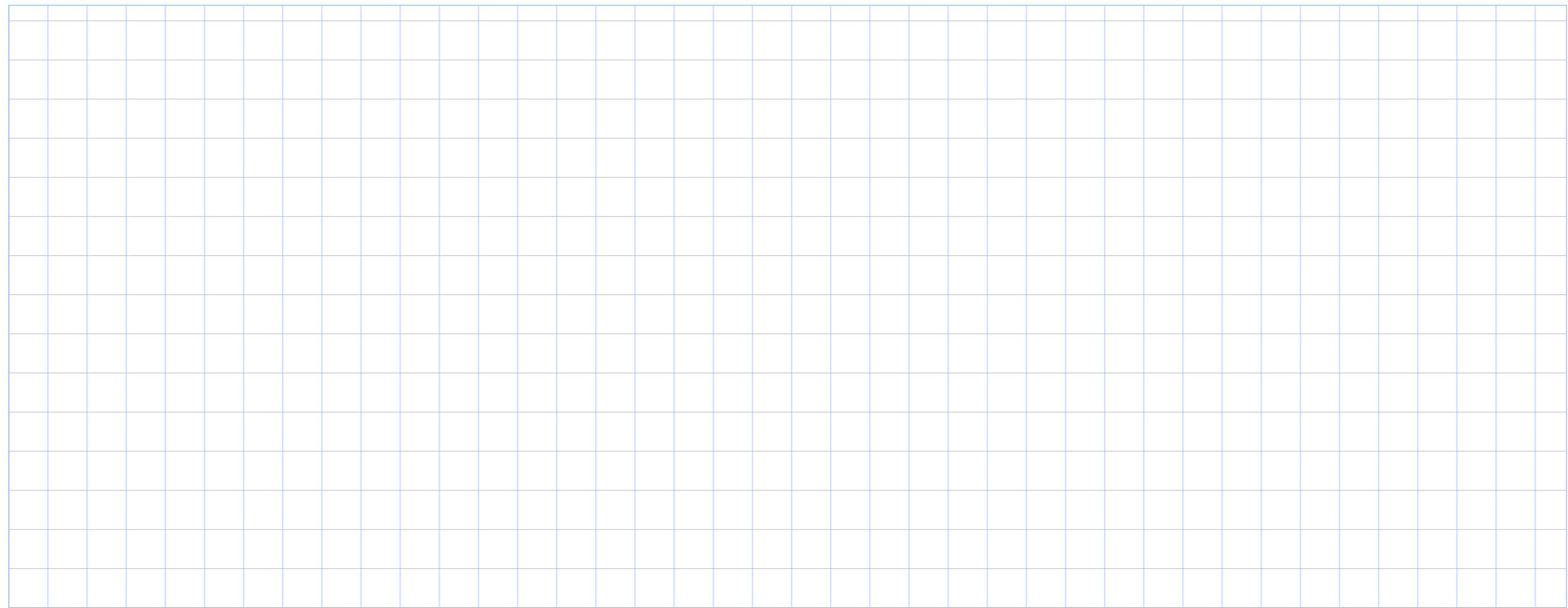
Relation

A **relation** is a set of ordered pairs. Its inputs form the **domain**; its outputs form the **range**.

Function

A **function** assigns *exactly one* output to every input. Different inputs may share the same output.

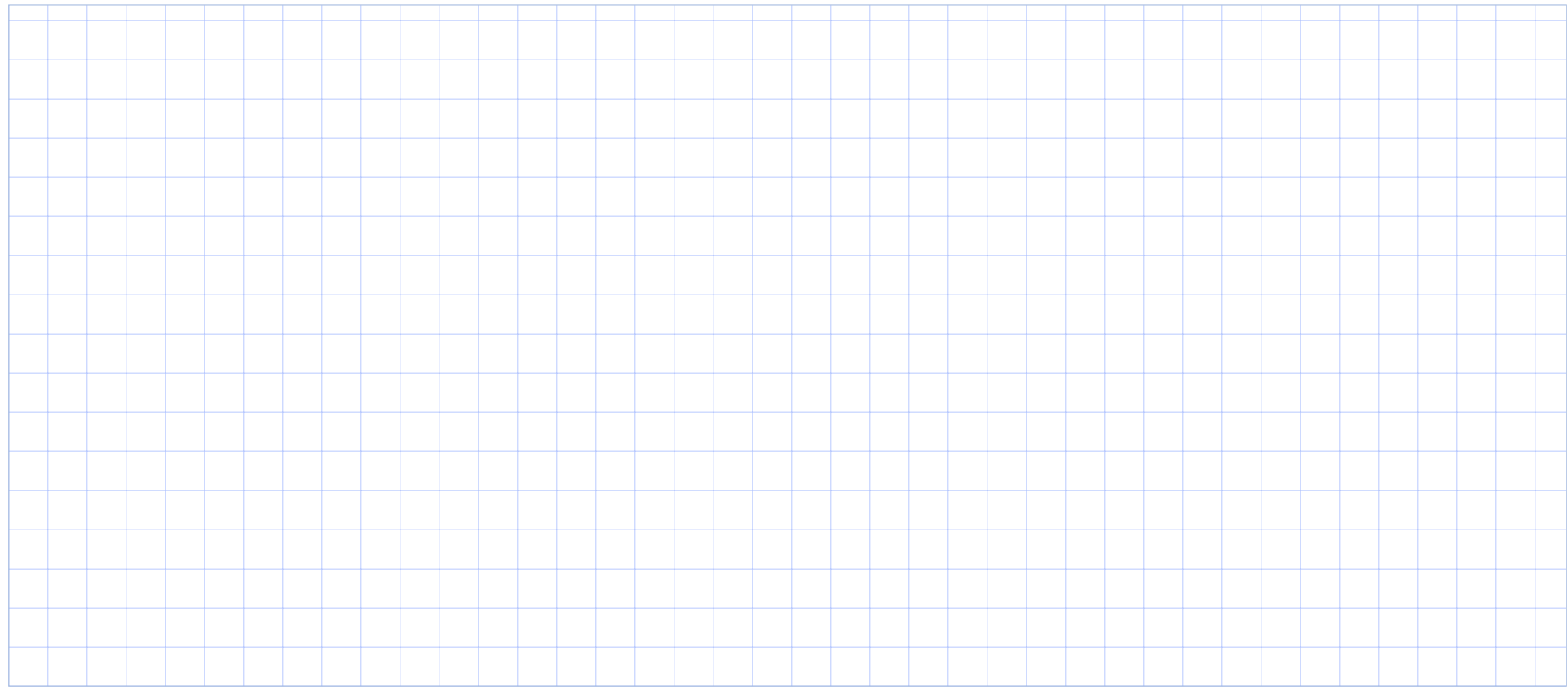
Function test: ask whether any single input is paired with more than one output. On a graph, this becomes the **vertical line test**.



Decide Whether Each Relation Is a Function

For each relation, give a reason—not only a yes/no answer.

- a. $\{(-2, 1), (0, 3), (2, 1), (0, -3)\}$
- b. $y^2 = x + 4$
- c. A graph passes through $(-3, 2)$, $(0, -1)$, $(2, 4)$, and $(5, 4)$, with no other points.

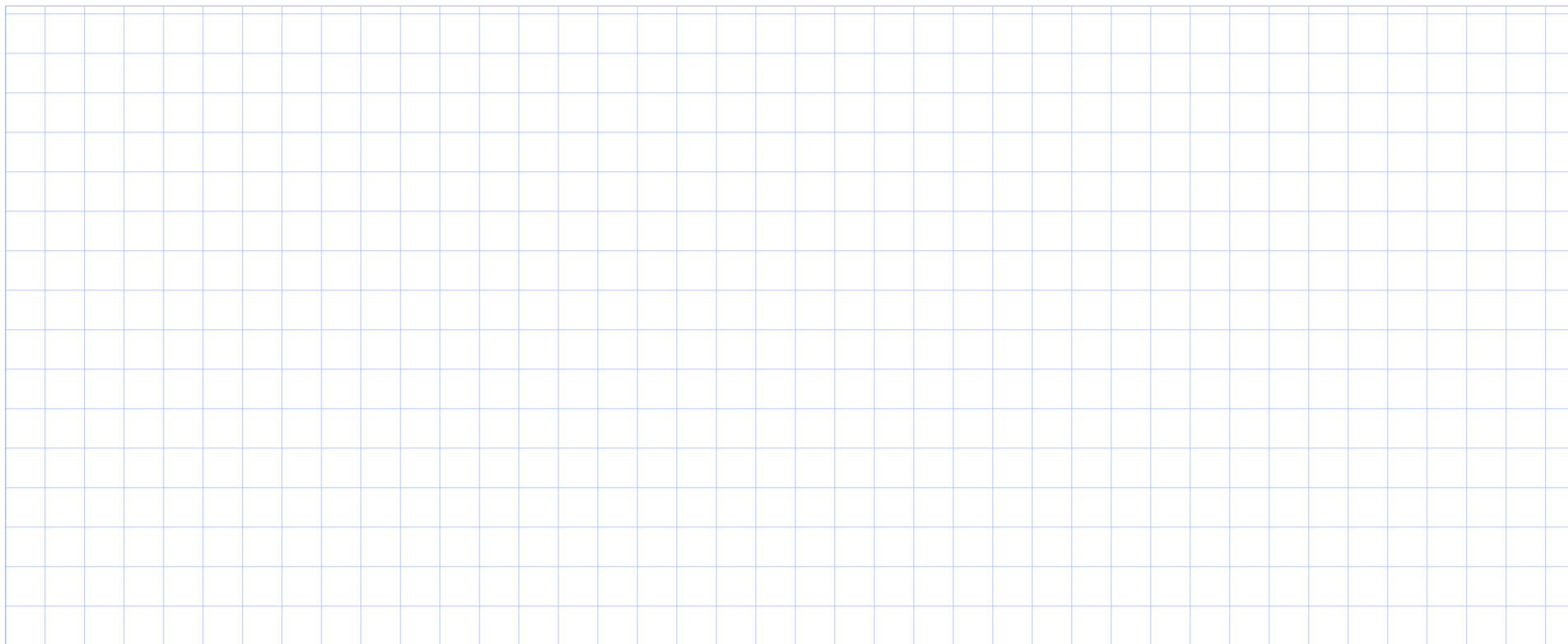


Find the Domain and Range

Let

$$f(x) = \frac{\sqrt{5-2x}}{x+1}.$$

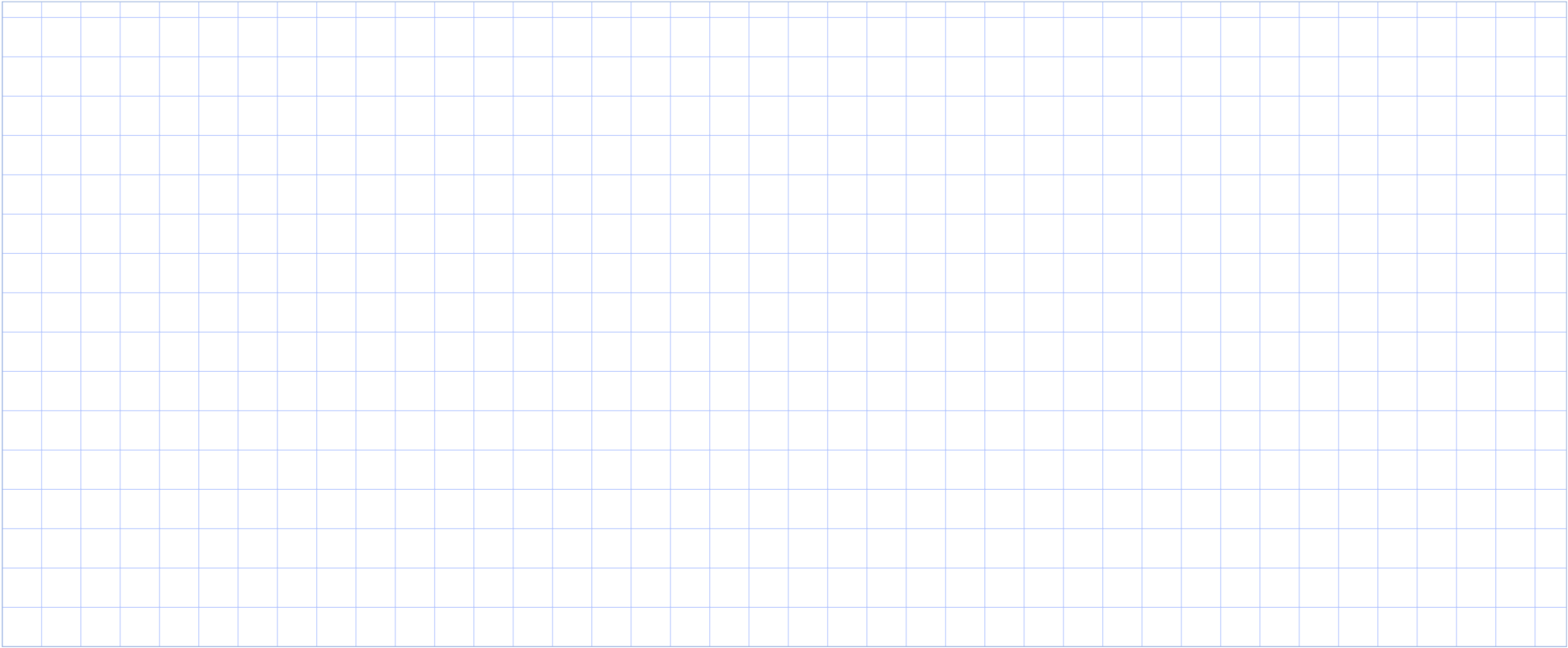
- State the domain in interval notation.
- Explain which restriction comes from the square root and which comes from the denominator.
- Without graphing technology, decide whether 0 belongs to the range. Justify your answer.



Combine Two Functions—Then Track the Domain

Let $f(x) = 2x - 3$ and $g(x) = \sqrt{x + 4}$.

- a. Find $(f \cdot g)(x)$ and state its domain.
- b. Find $\left(\frac{f}{g}\right)(x)$ and state its domain.
- c. Why are the two domains different even though the same functions are used?



Simplify a Difference Quotient

For $f(x) = x^2 - 3x + 2$, simplify

$$\frac{f(x+h) - f(x)}{h}, \quad h \neq 0.$$

Your final expression should not have h in the denominator. Clearly show where the common factor of h appears.

Zeros and intercepts

A real number r is a **zero** of f when $f(r) = 0$. Then $(r, 0)$ is an x -intercept. The y -intercept, if it exists, is found from $f(0)$.

Reading coordinates

$f(a) = b$ means the point (a, b) lies on the graph. Read an input horizontally and its output vertically.

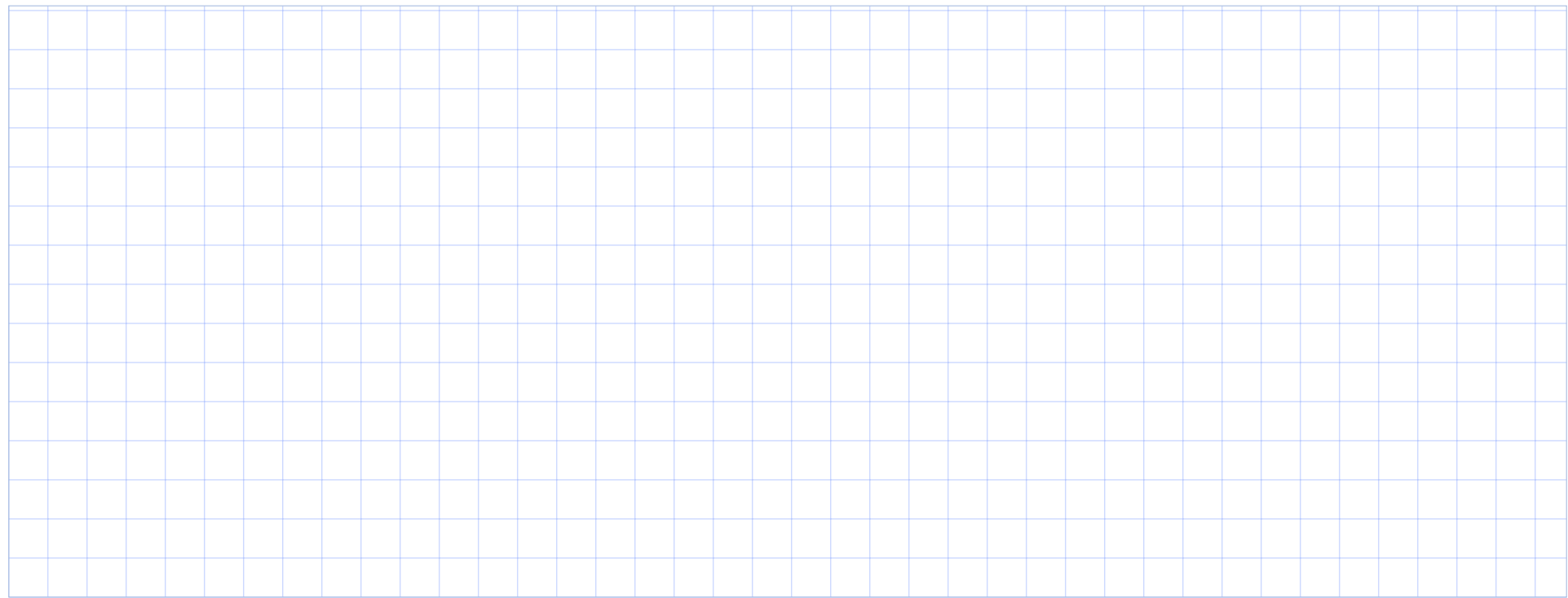
A graph may also reveal domain, range, symmetry, and how often a horizontal or vertical line intersects it. Endpoints and open/closed points matter.



Let

$$h(x) = \frac{x^2 - 4}{x - 1}.$$

- a. Is $(2, 0)$ on the graph of h ? What about $(-2, 0)$?
- b. Find the x - and y -intercepts.
- c. State the domain in interval notation.
- d. Evaluate $h(3)$ and interpret the result as a point on the graph.



Read a Piecewise Graph

Use the graph to answer:

- a. $f(-4)$, $f(0)$, and $f(3)$
- b. all x - and y -intercepts
- c. the domain and range
- d. all x such that $f(x) = 2$
- e. how many times $x = 1$ intersects the graph

